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Studierichting: Informatica
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Opgave: zoek $\frac{dw}{dt}$: $w = \frac{1}{u^2 + v^2}$ met $u = \cos 2t$ en $v = \sin 2t$

$$w = (u^2 + v^2)^{-1}$$

$$\Rightarrow \frac{\partial w}{\partial u} = -1 \cdot (u^2 + v^2)^2 \cdot 2u = \frac{-2u}{(u^2 + v^2)^2}$$

$$\Rightarrow \frac{\partial w}{\partial v} = -1 \cdot (u^2 + v^2)^2 \cdot 2v = \frac{-2v}{(u^2 + v^2)^2}$$

$$\Rightarrow \frac{du}{dt} = \frac{d}{dt}(\cos 2t) = -2 \sin 2t$$

$$\Rightarrow \frac{dv}{dt} = \frac{d}{dt}(\sin 2t) = 2 \cos 2t$$

$$\frac{dw}{dt} = \frac{\partial w}{\partial u} \cdot \frac{du}{dt} + \frac{\partial w}{\partial v} \cdot \frac{dv}{dt}$$

$$= \frac{-2u}{(u^2 + v^2)^2} \cdot (-2 \sin 2t) - \frac{2v}{(u^2 + v^2)^2} \cdot (2 \cos 2t)$$

$$= \frac{4 \cos 2t \cdot \sin 2t}{(\cos^2 2t + \sin^2 2t)^2} - \frac{4 \sin 2t \cdot \cos 2t}{(\cos^2 2t + \sin^2 2t)^2}$$

$$= 4 \cos 2t \cdot \sin 2t - 4 \sin 2t \cdot \cos 2t = 0$$

$$\frac{dw}{dt} = 0$$

References